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Differentially private de-identifying textual medical document is compliant with challenging NLP analyses: Example of privacy-preserving ICD-10 code association

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ABSTRACT

Medical research plays a crucial role within scientific research. Technological advancements, especially those related to the rise of machine learning, pave the way for the exploration of medical issues that were once beyond reach. Unstructured textual data, such as correspondence between doctors, operative reports, etc., often serve as a starting point for many medical applications.

However, for obvious privacy reasons, researchers do not legally have the right to access these documents as long as they contain sensitive data, as defined by regulations like GDPR (General Data Protection Regulation) or HIPAA (Health Insurance Portability and Accountability Act). De-identification, meaning the detection, removal or substitution of all sensitive information, is therefore a necessary step to facilitate the sharing of these data between the medical field and research. Over the past decade, various approaches have been proposed to de-identify medical textual data. However, while entity detection is a well-known task in the natural language processing field, it presents some specific challenges in the medical context. Moreover, existing substitution methods proposed in the literature often pay little attention to the medical relevance of de-identified data or are not very resilient to attacks.

This paper addresses these challenges. Firstly, an efficient system for detecting sensitive entities in French medical data and then accurately substitute them was implemented. Secondly, robust strategies for generating substitutes that incorporate the medical utility of the data were provided, thereby minimizing the difference in utility between the original and de-identified data, and that mathematically ensure privacy protection. Thirdly, the utility of the de-identification system in a context of ICD-10 code association was evaluated. Finally, various systems developed to tackle ICD-10 code association were presented while providing a state-of-the-art model in French.

1. Introduction

Artificial Intelligence (AI) is widespread in diverse fields encompassing finance, transportation, information management, and many others (to name but a few). The sphere of healthcare is by no means an exception to this ubiquitous trend. This influence even extends to the manipulation of unstructured data, such as textual information. Prediction that were once insurmountable are now seen as achievable challenges. These include tasks such as the identification of similar patient records, automatic ICD labelization (Amin et al., 2019; Mahdi, Deligiannis, & Sahli, 2023; Névéol et al., 2018; Ponthongmak et al., 2023; Tchouka, Couchot, Laiymani, Selles and Rahmani, 2023), the anticipation of hospital re-admissions (Agarwal, Baechle, Behara, & Zhu, 2017; Thapa, Seifollahi, & Taheri, 2022), healthcare-associated infections detection (Scardoni, Balzarini, Signorelli, Cabitza, & Odone, 2020; Tvardik et al., 2018), to name but a few.

Nonetheless, mainly due to resource and time constraints within healthcare institutions, the execution of these processes requires the involvement of computer experts specialized in deep learning, data science, and big data analysis. Consequently, it is becoming imperative to facilitate data exchange between healthcare practitioners and data science specialists. Given the sensitive and critical nature of the data in question, such collaboration requires a rigorous de-identification

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procedure, which can only be carried out within a legal framework governing the healthcare stakeholders. Notable examples of institutional regulations include the U.S. Health Insurance Portability and Accountability Act (HIPAA) (Cohen & Mello, 2018) and the European General Data Protection Regulation (GDPR) (EU, 2016).

These regulations specify that data anonymization is a prerequisite for data sharing. When processing tabular data, each record contains the values of a set of collected attributes. Numerous methods have addressed this or that type of attribute (Ding, Kulkarni, & Yekhanin, 2017; Erlingsson, Pihur, & Korolova, 2014) and their combinations in an array (Sweeney, 2002; Wang et al., 2019). However, when it comes to unstructured data, the problem of de-indentifiability is more complex, as one has to work with a set of a priori unorganized information. But this information is in fact organized (the words form sentences) and this organization must be preserved. Indeed, medical information after de-identification must be as rich as it was before de-identification to hope for subsequent processing by AI, as in the examples mentioned above.

This study focuses on the ability to exchange textual medical documents, often written by healthcare professionals, which can take the form of surgical reports, clinical notes, or biological test results.

To enhance privacy protection, de-identification techniques (Bourdois et al., 2021; Dernoncourt, Lee, Uzuner, & Szolovits, 2016; Hartman et al., 2020; Hassan, Sánchez, & Domingo-Ferrer, 2021; Jordan, Fontaine, & Hendricks-Sturrup, 2022; Lison, Pilán, Sánchez, Batet, & Øvrelid, 2021) have been presented as a means to mask or substitute any form of Protected Health Information (PHI) attributed to a patient, making it difficult to associate a person with their specific data. The components constituting PHI are in part defined by the privacy regulations in force within the jurisdiction concerned. As an illustrative example, HIPAA provided 18 distinct categories of PHI, encompassing designations such as names, geographic locations, and telephone numbers. In the European context, given that GDPR does not explicitly offer PHI definitions, the majority of researches adhere to the stipulated HIPAA criteria.

The de-identification process is generally described as an algorithm comprising two main steps, namely first, the identification of all sensitive information, and, secondly, the replacement of these identified elements.

The initial step encompasses the automated detection of entities within a given text sequence. In the realm of Natural Language Processing (NLP), this phase is commonly referred to as Named Entity Recognition (NER). However this task presents several unique challenges in a medical context, one of which is the scarcity of a consistent training dataset in the specific language relevant to the effective model development (which is essentially the dataset that the de-identification task seeks to generate). Another notable challenge lies in the need to refine and adapt generic natural language processing models to the specialized lexicon and terminology prevalent in the field of medicine.

The subsequent step involves the substitution of the identified sensitive entities. Unfortunately, this step is frequently overlooked in research efforts, as it is perceived as less critical. This omission is particularly evident when dealing with entities such as phone numbers or email addresses, which can be replaced by arbitrary random values, given their lack of direct correlation with medical information. However, the same cannot be said for entities such as dates or geographic locations. For example, the analysis of the medical file of a patient under 18 years of age is very different from that of a patient over 50. Similarly, the patient's location, whether in an area with high pollution levels or a region 7

7 with low pollution, has notable implications for the medical assessment. The challenge in this second step lies in striking a delicate balance between safeguarding individual privacy and preserving the medical utility of the data, which remains essential for subsequent machine learning tasks.

This study is motivated by the objective of developing a Machine Learning approach aimed at assigning numerical codes from the International Classification of Diseases, 10th Revision (ICD-10), to medical documents such as patient records, medical reports, and clinical notes. The ICD-10 is a globally recognized standard classification system that assigns numeric codes to various medical conditions, symptoms, medical procedures, and other health-related information. The task of associating ICD-10 codes with medical documents presents several significant challenges, including the large number of codes to be classified, the scale of the input data, language variations (e.g., French versus English) in the automatic natural language processing models, imbalanced datasets, and many others.

However, to develop this Machine Learning approach, access to data is necessary, and this is where the entire de-identification process becomes significant: enabling prototyping on data sanitized of identifying information while preserving a substantial portion of medical information.

This article constitutes a comprehensive reworking of articles (Tchouka et al., 2022; Tchouka, Couchot and Laiymani, 2023; Tchouka, Couchot, Laiymani, Selles et al., 2023), demonstrating the feasibility of establishing a comprehensive process for de-identifying medical notes based on Local Differential Privacy. In Table 1 we summarize the evolution of our works and for this paper our contributions can be summarized as follows:

1. We developed a sensitive French Named Entity Recognition system based on the Transformer architecture (FlauBERT model). This model is trained on a constructed dataset and represents the state-of-the-art in named entity recognition in the French language within the context of de-identification with HIPAA attributes. An open-source implementation of this NER task in available on GitHub[1]
2. We formalized robust strategies for generating substitutes for sensitive attributes based on Local Differential Privacy based on a metric (metric-privacy) that ensures security and preserves medical utility. An open-source implementation of the surrogate generation approaches provided in is available on GitHub.[2]
3. These first two contributions enabled the development of an incremental approach for dataset construction: from a partially manually labeled dataset of 375 individual records to a de-identified dataset containing over 46,000 documents, in accordance with the criteria outlined earlier.
4. We developed various architectures to tackle the issues of ICD coding, namely the significant number of labels and the size of medical data. The proposed "CamemBERT+LAAT" architecture represents the state-of-the-art in the French language for ICD-10 code association task. An open-source implementation of this system is available on GitHub.[3]
5. We evaluated our de-identification approach in the context of ICD-10 code association task. Our de-identification approach helps reduce the loss of utility (6.8%) compared to a traditional de-identification method (12%, e.g., replacing sensitive attributes with their labels).

This work is conducted in collaboration with the Hopital Nord Franche-Comté (HNFC), a French public health center that generously provided us with patient records. To uphold privacy standards, all experiments were conducted on-site, and no data was extracted from the hospital premises.

This paper is organized as follows. Section 2 presents the state of the art in the various domains we have addressed, namely, Natural Language Processing, de-identification, and the association of ICD-10

[1] https://github.com/mlfiab/ner-french.
[2] https://github.com/mlfiab/surrogate-deid.
[3] https://github.com/mlfiab/icd10-french.

Table 1
Comparison of our main contributions in articles (Tchouka et al., 2022; Tchouka, Couchot and Laiymani, 2023; Tchouka, Couchot, Laiymani, Selles et al., 2023).

		Tchouka et al. (2022)	Tchouka, Couchot and Laiymani (2023)	Tchouka, Couchot, Laiymani, Selles et al. (2023)	Proposal
De-identification	NER	*Hybrid method*	*Deep learning only*	*NA*	*Deep learning only*
		MEDINA + FlauBERT	New labeled dataset FlauBERT		Rewriting Publication on GitHub
	NES	*ϵ-LDP based*	*ϵ-metric privacy based*	*NA*	*ϵ-metric privacy based*
		Date: bounded Laplace mechanism	Date: metric Laplace mechanism		Rewriting & formalizing
		Location: GeoIndisdinguishability	Location: exponential mechanism		Publication on GitHub
ICD-10 classification	Datasets	*128 documents*	*NA*	*ICD_10 HNFC*	*De-identified dataset*
		64 original ones		56,000 documents	ICD_10_DEID_HNFC ICD_10_TAG_HNFC
		64 de-identified ones		With ICD-10 associated codes	With ICD-10 associated codes
	Evaluation & models	*Manual task*	*NA*	*New models*	*Models improved*
		ICD-10 association on all documents		CamemBERT + LAAT	Better de-identified dataset
					Rewriting of Dalloux, Claveau, Cuggia, Bouzillé, and Grabar (2020) model
		Manual confusion matrix computation		Bibliographic comparison with Sota	Fair experimental comparison

codes. Subsequently, we discuss in Section 3 the de-identification with the Named Entity Recognition task (Section 3.1), followed by substitution (Section 3.2). Next, in Section 4, we delve into the application context of this paper, the association of ICD-10 codes. Then, Section 5 presents the methodology to measure the utility of de-identification in the context of ICD-10 code association. Section 6 presents all the experiments conducted within the scope of this work. Finally, Section 7 discusses the relevance of our approaches in the time of LLMs and Section 8 presents our concluding remarks.

2. Related work

In this section, we present the state of the art in the various domains we have addressed, namely, Natural Language Processing, de-identification, and the association of ICD-10 codes.

2.1. Natural language processing

NLP has made significant advances in recent years, mainly thanks to the introduction of the Transformers model (Vaswani et al., 2017). These models feature a remarkable capacity for transfer learning, as demonstrated by the BERT model (Devlin, Chang, Lee, & Toutanova, 2018), which proved their effectiveness in providing more accurate contextualized representations. Subsequently, a range of pre-trained models emerged, including BERT, RoBERTa (Liu et al., 2019), and others. These models undergo pre-training on an extensive general domain of a given language text to capture text data modeling capabilities. They are then refined for specific classification tasks.

In the field of French language processing, two prominent models have been introduced: FlauBERT (Le et al., 2019) and CamemBERT (Martin et al., 2020). Additionally, there are multilingual models like XLM-R (Conneau et al., 2019). Some models specialize in domain-specific text, ClinicalBERT (Alsentzer et al., 2019) and BioBERT (Lee et al., 2020) for instance, which are trained on medical data for medical domain tasks. Notably, there is a gap in French language models, which hinders the application of machine learning approaches to French documents compared to English ones. In the French language, we can mention DrBERT, introduced recently by Labrak et al. (2023), trained on a large corpus of medical data in the French language and based on the BERT architecture.

Transformers models generally have a limited input size, typically around 512 tokens in practice. However, this limitation becomes especially problematic for clinical documents, which are often much larger than 512 words or tokens. In addressing this challenge, a hierarchical approach was proposed in Pappagari, Zelasko, Villalba, Carmiel, and Dehak (2019). The document is segmented, and each segment is processed by a Transformers model. The encoded segments are then aggregated in the next layer, using techniques such as linear layers, recurrent neural networks, or other layers of Transformers. In recent years, a sparse-attention system known as the *LongFormer* model has been introduced in Beltagy, Peters, and Cohan (2020). This model incorporates both local attention (within a window of neighboring tokens) and global attention, effectively reducing the computational complexity. Consequently, the LongFormer model can handle document sizes of up to 4096 tokens.

Recently, NLP has experienced a major breakthrough with the introduction of new learning approaches using Large Language Models (LLM), such as Reinforcement Learning from Human Feedback (RLHF) (MacGlashan et al., 2017) on GPT (Radford et al., 2019). Thanks to these LLM, training techniques such as zero-shot learning (Agrawal, Hegselmann, Lang, Kim, & Sontag, 2022) and few-shot learning (Su et al., 2022) make it easier to approach NLP tasks.

2.2. De-identification

2.2.1. Named entity recognition

Detecting entities in a text sequence is a well-established task in automatic natural language processing, commonly referred to as Named Entity Recognition (NER). Initially, rule-based systems were used, in which a set of rules covered all the potential syntaxes of the attributes to be identified. With the advent of machine learning, more precise systems emerged, using models such as SVM, Decision trees, or Conditional Random Field (CRF). The transition to neural networks saw the introduction of deep learning models by researchers like Dernoncourt

et al. (2016) and Liu, Tang, Wang, and Chen (2017). Dernoncourt's recurrent neural networks achieved impressive F_1-scores of 99.23% and 97.85% on i2b2 (Sun, Rumshisky, & Uzuner, 2013) and MIMIC (Johnson et al., 2016) datasets, respectively, while taking into account attributes outlined by the HIPAA law (Cohen & Mello, 2018).

With the rise of transformers (Vaswani et al., 2017) and BERT in NLP, which excel in contextualized text representation, it has become obvious that transformer-based models are the most accurate for NER. Notable works in this field include Hanslo (2021) and Polignano, de Gemmis, and Semeraro (2021). In a recent study (Liu et al., 2023), GPT-4 with the "zero-shot learning" technique (Pourpanah et al., 2022) was employed for NER in the context of de-identification, achieving a remarkable F_1 score of 99% on the i2b2 dataset. In comparison, models using BERT and ClinicalBERT achieved 79.8% and 97.4%, respectively.

Research on the de-identification of French medical documents, led mainly by Grouin, Griffon, and Névéol (2015), used a CRF-based machine learning model that achieved an 80% F_1-score. Recent work (Bourdois et al., 2021) focuses on de-identifying French emergency medical records. It uses a two-step approach: the FlauBERT model assigns labels to documents requiring de-identification, followed by a combination of rules-based techniques and LSTM through Flair (Akbik et al., 2019).

Unfortunately, there is no French equivalent to datasets like MIMIC or i2b2. This led authors in Tchouka et al. (2022) to combine MEDINA a CRF based machine learning method (Grouin et al., 2015) and a neural network method based on transformers on the WikiNER dataset (Nothman, Ringland, Radford, Murphy, & Curran, 2013) to integrate all attributes for detection. This hybrid system achieves a baseline F_1-score of 94.7% in this context.

Note that the use of multilingual models such as XLM-Roberta (Conneau et al., 2019) can be an interesting approach when the datasets are scarce. In Silvestri, Gargiulo, Ciampi, and Pietro (2020) the authors show that such models can be useful. Nevertheless, in Kanaan et al. (2023) this approach was tested on a similar problem of classifying emergency phone calls and the obtained results were not as good as the full French approach i.e. CamemBERT or FlauBERT with a small French dataset.

2.2.2. Entity substitution

Once the entities have been detected, they need to be cleaned up - this is the substitution phase. The simplest method consists in replacing the detected information by its entity (the name "Durant" by the entity "NAME"). While this protects the privacy of individuals, it degrades the structure, readability and coherence of the document. For this reason, more coherent de-identification methods have emerged (Deleger et al., 2014; Stubbs, Uzuner, Kotfila, Goldstein, & Szolovits, 2015; Uzuner, Luo, & Szolovits, 2007). Sensitive information is replaced by random information that preserves the document's structure (the name "Durant" is replaced by a random name "Julien", or a date by a date shifted by a few days, etc.). In these strategies, the aim is to preserve the document's structure and information while protecting privacy. In the medical context, the substitution task can become complex to carry out. In general, in medical analyses, some of the sensitive attributes to be anonymized may have an impact on document analysis, such as ages, dates or geographical locations.

In Tchouka et al. (2022), for temporal entities, the authors opted for Local Differential Privacy (Duchi, Jordan, & Wainwright, 2013) with bounds in time categories to calibrate the added noise. Regarding geographical locations, geo-indistinguishability (Bordenabe, Chatzikokolakis, & Palamidessi, 2014) was retained as a direct mechanism to provide a location close to the original one. While these two privacy-preserving methods safeguard the data privacy in an efficient way, it is important to note that they may introduce unnecessary additional noise in the temporal aspects, and in some cases, they could potentially substitute a large urban area with a small village. This substitution may result in a significant disparity in the epidemiological characteristics of the data, as the characteristics of a small village could be vastly different from those of a city.

In Hassan et al. (2021), the authors use ontology based generalization techniques such as YAGO (Fabian, Gjergji, Gerhard, et al., 2007) to substitute detected entities and security is observed through evaluating robustness against re-identification attacks. Finally, in articles (Carvalho, Vasiloudis, Feyisetan, & Wang, 2023; Feyisetan, Balle, Diethe, & Drake, 2020), the authors present an approach adding noise verifying ϵ metric-privacy to vector representation of words resulting from word embedding models. However, this one-step approach is not compliant with HIPAA constraints.

2.3. ICD code association

The automatic association of ICD codes stands as a prominent challenge in medical research, and with the advancements in neural networks and the evolution of natural language processing, numerous approaches have been explored. Authors such as Baumel, Nassour-Kassis, Cohen, Elhadad, and Elhadad (2018) and Choi, Bahadori, Schuetz, Stewart, and Sun (2016) used recurrent neural networks (RNNs) to encode Electronic Health Records (EHR) to predict diagnostic outcomes. Conversely, Mullenbach, Wiegreffe, Duke, Sun, and Eisenstein (2018) and Shi, Xie, Hu, Zhang, and Xing (2017) incorporated attention mechanisms with RNNs and CNNs to enhance the accuracy of the model.

Tsai, Chang, and Chen (2019) and Xie and Xing (2018) introduced diverse strategies to take into account the hierarchical structure of codes. Xie and Xing (2018) employed a sequence tree LSTM to capture the hierarchical relationships and semantics of each code, while Cao et al. (2020) proposed to train the integration of ICD codes in a hyperbolic space, leveraging a graph neural network to capture code co-occurrences. LAAT (Vu, Nguyen, & Nguyen, 2020) integrates a bidirectional LSTM with an attention mechanism incorporating labels.

EffectiveCAN (Liu, Cheng, Klopfer, Gormley, & Schaaf, 2021) introduced a squeeze-and-excitation network, residual connections, and representation extraction from all encoder layers for label attention. They addressed long-tail prediction issues with focal loss, achieving an F_1-score of 58.9% on MIMIC 3 (Johnson et al., 2016). ISD (Zhou et al., 2021) used shared representation extraction and a self-distillation learning mechanism to address the distribution of long-tailed codes with 55.9% of F_1-score.

Ponthongmak et al. (2023) proposes models based on neural networks using a supervised learning system. The authors developed convolutional neural network (CNN) with PubMedBERT embeddings, using a dataset of 15,329 medical documents. They validated their approaches on the MIMIC 3 dataset with an 55.65% F_1-score. Huang, Tsai, and Chen (2022) proposed the PLM-ICD system, focusing on document encoding with multi-label classification. They used a transformer-based encoding model adapted to medical corpora. To handle a large set of labels (e.g., MIMIC 3 with over 8000 codes), they used the Label-Aware Attention (LAAT) mechanism (Vu et al., 2020) and implemented a hierarchical method for long sequences. PLM-ICD is the current state-of-the-art model, achieving 59.8% F_1-score on MIMIC 3 (Johnson et al., 2016) and 50.4% on MIMIC 2 (Saeed et al., 2011).

In the French context, Dalloux et al. (2020) proposed Convolutional Neural Networks (CNN) models with multi-label classification system for ICD-10 code association. They used FastText vectors (Bojanowski, Grave, Joulin, & Mikolov, 2016) for document encoding, considering both the final labels and grouped families to reduce the class count. Trained on a private dataset of 28,000 clinical documents, their model achieved an F_1-score of 39% with 6116 codes and 52% with 1549 codes. In Mahdi et al. (2023), interested readers can find a summary of the state-of-the-art of the ICD coding models which was just mentioned. In Tchouka, Couchot, Laiymani, Selles et al. (2023), the authors used a multi-label classification system as in Huang et al. (2022). They experimented with various architectures based on transformers models

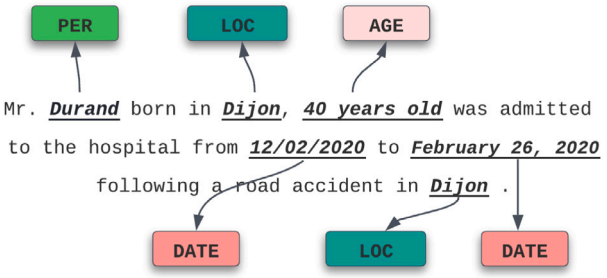


Fig. 1. Perfect NER of PHI entities on thread example.

Table 2
Description of sensitive attributes based on the HIPAA law.

Label	Description
PER	All names of persons
DATE	All date sequences in all formats
LOC	All geographical locations and zip codes
ORG	Organizational entities
AGE	Ages
TEL	Phone numbers
REF	All references related to individuals
QID	Any ID sequence

to address the challenges posed by a significant number of codes and the size of input sequences. The authors propose a code association system based on the most frequent codes. The most advanced model from Tchouka, Couchot, Laiymani, Selles et al. (2023) currently represents the state-of-the-art in the task of associating ICD-10 codes in the French language.

3. De-identification system

In this section, the de-identification process is discussed. Both de-identification stages (named entity recognition and entity substitution) are presented as well as our proposals for each of them. The following is our guiding example that will be developed throughout this section.

Thread Example. Consider the following fictional sentence. It is typical of what can be found in the medical text of a hospital. "Mr. Durand born in Dijon, 40 years old, was admitted to the hospital from 12/02/2020 to February 26, 2020 following a road accident in Dijon".

3.1. Named entity recognition

Section 3.1.1 discusses the motivation for the need to strengthen the NER step. In Section 3.1.2, the methodology to build the training dataset is presented. Finally, in Section 3.1.3, details are provided on the architecture of our NER model and the training process.

Thread Example. Fig. 1 illustrates the result of a perfect detection (NER) process applied to the threaded example. HIPAA labels (Cohen & Mello, 2018) with their descriptions are summarized in Table 2.

3.1.1. Motivation and global overview

Achieving near-perfect scores in Named Entity Recognition (NER) is crucial to the success of the de-identification process. Undetected sensitive information presents a risk of document re-identification. NER tasks are problem-dependent, often lacking the ideal dataset for specific issues. The most successful NER results in de-identification, as seen in the literature, have been achieved by implementing English models on domain-specific datasets that are both consistent and comprehensive. For the French language, datasets with sufficient data to build an efficient model are rare, and to the best of our knowledge, none exists

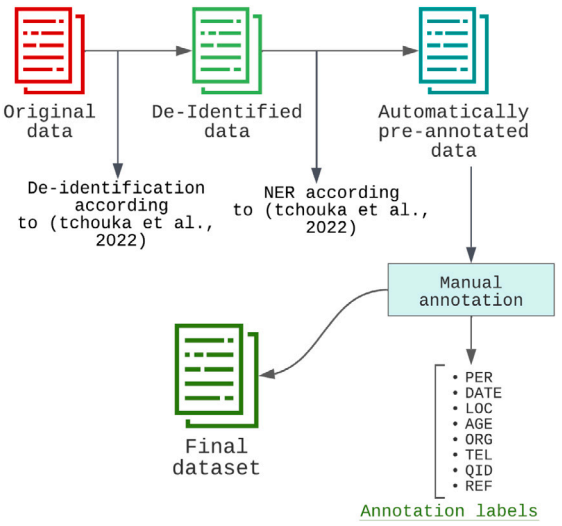


Fig. 2. FrenchHospitalNER dataset construction.

in the medical domain. Consequently, it becomes imperative to create a suitable dataset tailored to our application's context—a medical corpus containing all relevant identifying attributes. With access to such a dataset, a Transformer-based NER method can be leveraged.

3.1.2. Building a labeled dataset

As mentioned, the most challenging step is to find a sufficiently large dataset to implement an accurate model that encompasses all categories of sensitive information and is adapted to the medical corpus, especially in French. Unfortunately, such a dataset is currently unavailable or not easily accessible to everyone. In Tchouka et al. (2022), the authors faced similar challenges, using the WikiNER dataset, which had only a few tags and a very general vocabulary. To address this limitation, they had to use multiple methods to integrate all categories into a de-identification tool. In this work, carried out in collaboration with a French public hospital, we have on-site access to a substantial set of unlabeled medical notes. Our proposal involves members of the hospital to semi-manually annotate a subset of these notes.

The process is illustrated in Fig. 2. Given that there may be a necessity to access these documents in subsequent stages, this entity recognition process begins with the de-identification step described in the document in Tchouka et al. (2022). Consequently, an automated pre-annotation phase is executed using the same tool presented in Tchouka et al. (2022). This step is purely for efficiency's sake. This is followed by a third phase of human verification, correction, and enhancement of these annotations, which constitutes the most crucial step of this process. Thanks to this pre-annotation phase, the entire annotation process only took 25 h for one individual (equivalent to 1 min per file). The ensuing dataset, labeled as *FrenchHospitalNER*, is thus de-identified according to the methodology described in Tchouka et al. (2022) and includes tags suitable for NER based on the Transformer model. It contains 14,925 sentences.

3.1.3. Supervised learning on a dedicated labeled dataset

This section starts with the introduction of our model's architecture. Subsequently, the implementation details of supervised learning are examined in depth. Finally, an evaluation of our modelis given, drawing comparisons with existing de-identification models.

Model architecture: Transformer based approach

The accessibility of FlauBERT, a BERT-based pre-trained French model, has influenced our decision to use this transformer. This choice is further supported by findings in Polignano et al. (2021), demonstrating that language-specific models for French, such as FlauBERT, yield improved results compared to multilingual BERT models.

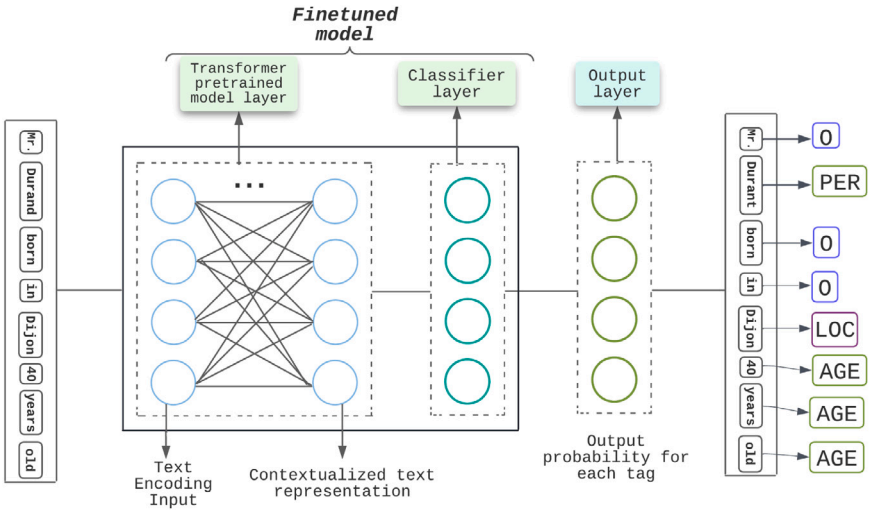


Fig. 3. Deep learning model architecture for NER.

Fine-tuning transformer model

Starting with a pre-trained model like FlauBERT, the remaining task is to refine it on a smaller and more specialized dataset. Instead of creating our text classification or feature detection model from scratch, our approach is to start with the pre-trained BERT and add a classification layer i.e. a fully-connected layer. This is the transfer learning process introduced by Howard and Ruder (2018). This process is illustrated in Fig. 3.

Training

The learning process has been executed using the aforementioned dataset, employing a deep learning-based NLP model as described earlier.

- the **Learning Rate** controls the size of the update steps along the gradient. Usually, a very small value is set (10^{-4} in this work), so that the weights are less modified at each iteration, which avoids missing the optimal values of the error function
- the **Dropout** is a regularization technique for reducing overfitting in neural networks. It is set to 0.1 in this work, which means that 10% of selected neurons are ignored during training
- the **Training Batch Size** is the number of training samples to work through before the model's internal parameters are updated.
- the **Maximum length** defines the maximum number of words in the sentences
- the **Number of epochs** is the number of complete passes through the training dataset.

The Named Entity Recognition (NER) model is formulated as a multi-class classification task, where tags serve as classes. The CrossEntropy error function (De Boer, Kroese, Mannor, & Rubinstein, 2005) is well-suited for this purpose. For optimization through back-propagation, the AdamW algorithm (Loshchilov & Hutter, 2017), one of the latest advancements in optimizers known for its effectiveness in neural network learning, isused. The FrenchHospitalNER dataset is randomly divided into training and test sets at a ratio of 90/10.

In the realm of machine learning, the challenge lies in determining the optimal values for hyper-parameters to achieve the most accurate results. Various methods exist for hyper-parameter optimization, including Grid or Random Search, model-based Bayesian methods... Studies, such as Bergstra, Yamins, and Cox (2013), indicate that Bayesian methods tend to yield more accurate results. In this paper, the Tree-structured Parzen Estimator (Bergstra, Bardenet, Bengio, & Kégl, 2011), a classical Bayesian optimization algorithm suitable for a classification model like ours, was used for hyper-parameter optimization. The experiments were conducted with different parameter

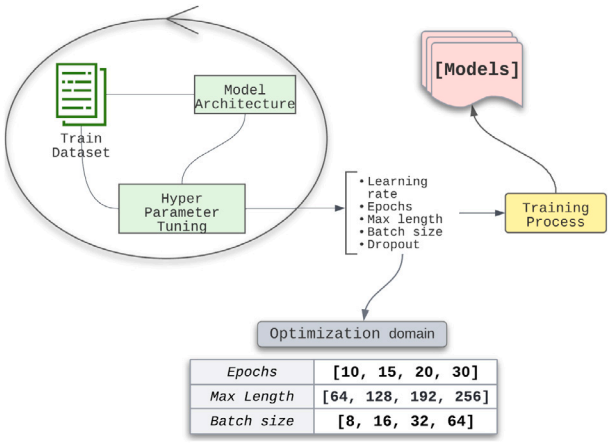


Fig. 4. Hyper-parameter optimization process. Search domains are provided for number of epochs, maximum length, and batch size.

combinations based on the Tree-structured Parzen Estimator algorithm, as illustrated in Fig. 4.

The training optimization process focused on three key hyper-parameters: the number of epochs, the maximum text length, and the batch size. Among all the hyper-parameter configurations, the one with the highest F_1 score is *number of epochs* = 20, *maximum length* = 128, and *batch size* = 64. The model with this configuration is then used in the next Evaluation section.

3.1.4. Evaluation

To evaluate our model we used the classical metrics Precision (P), Recall (R), and F_1-score (van Rijsbergen, 1979), which is an harmonic mean of both.

To get a sense of the overall performance of the system, the micro-average system was used. To be fair with Tchouka et al. (2022), HNFC hospital provides the same validation set already presented in Tchouka et al. (2022) that will refer to as HNFC dataset.

HNFC dataset contains 375 documents of deceased patients from HNFC hospital. These documents were pre-annotated by the hybrid system proposed by Tchouka et al. (2022) and then were manually annotated by the hospital staff with Doccanno manual annotation tool (Nakayama, Kubo, Kamura, Taniguchi, & Liang, 2018). Given the automatically pre-annotated files, the annotator is responsible for checking, correcting and completing the possible errors produced by

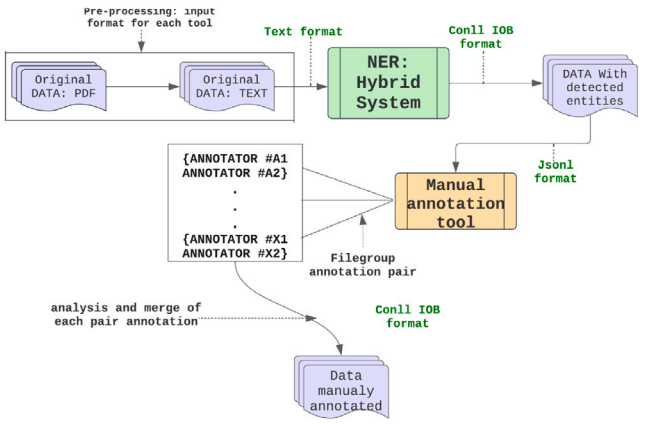


Fig. 5. HNFC dataset construction.

the model. HIPAA attributes were formalized to avoid ambiguity and multiple annotation criteria. To minimize the risks, two annotators worked in parallel on the same files and each annotation pair was then manually analyzed and merged into an unique one. This work was carried out by 6 people over 6 h. This process is illustrated in 5.

The i2b2 dataset (Sun et al., 2013) contains over 20,000 medical labeled reports enabling the examination of diverse biomedical topics such as the automatic assignment of ICD-10 codes (as discussed later). When used by Dernoncourt et al. (2016), only 40% of it is taken into consideration.

To establish a baseline, various existing entity detection solutions were evaluated on the same evaluation dataset. Here are the baseline models that were integrated into our study:

- CamemBERT-ner (Polle, 0000) is a NER model that was refined from camemBERT (Martin et al., 2020) on WikiNER dataset (Nothman et al., 2013),
- MEDINA (Grouin et al., 2015) is a method that uses CRF to label entities such as dates, phone numbers, email addresses, and URLs in medical texts written in French,
- FlauBERT-ner: A model built on deep learning from a pretrained transformers model, FlauBERT, also trained on Wiki-NER.

The results are detailed in Table 3. In this table, the highest scores in any language (resp. in French language only) for each entity (PER, ORG, LOC...) with respect to Precision, Recall and F_1 metric are marked in **bold**, (resp. in *italic*). In French, this proposal significantly outperforms all other approaches for almost all entities and for the micro-average aggregation. The enhanced performance can be attributed to the incorporation of a BERT-based layer, enabling precise contextualization of the sequence. Compared to the English evaluation (on two distinct datasets), the results of this proposal are not so far from those presented in Dernoncourt et al. (2016), which are recognized as the state of the art. Our comparatively lower score in the organization category, when contrasted with the i2b2 model, can be attributed to the informal structuring of organizations in medical documents, often in the form of abbreviations or isolated words. Increasing the dataset is expected to alleviate those difficulties and contribute to overall improvements of the scores in the different categories.

The next step consists in substituting the detected entities, as described in the next section.

3.2. Surrogate generation strategies

The challenge here lies in replacing personally identifiable information (PII) detected by Named Entity Recognition (NER) with relevant surrogates related to medical content while preserving privacy.

When it comes to preserving medical information, not all the entities need the same level of attention. For instance, replacing a phone number by any other random phone number does not change the medical information. On the othe hand, replacing location and date entities with random ones may lead toa loss of geographic epidemiological data for the former and chronological aspect for the latter.

As for the name elements, a random and memoïzation (Arcolezi, Couchot, Al Bouna, & Xiao, 2022; Erlingsson et al., 2019) based algorithm preserving the affiliation within the documents is provided in Tchouka et al. (2022) and is kept there.

Thread Example. In the thread example, "Durand" could be replaced by any name, for example, "Julien".

Time and location data on the other hand inherently carry information that is both medically critical and highly identifiable. This work will focus on these attributes, sanitized with Local Differential Privacy based approach first recalled hereafter. Local Differential Privacy (Duchi et al., 2013) formalizes the algorithm's robustness, and its definition is recalled in next section.

3.2.1. Local Differential Privacy (LDP)

An algorithm adhering to the principles of Local Differential Privacy (LDP) should be applied before granting access to the data curator, either by the data owner him/herself or as a pre-treatment input for the data curator.

Such an algorithm ensures that when data owners publish a sanitized output, it does not modify at all (from a probabilistic point of view) the ability to distinguish between them and other data owners. As a consequence, the wider the range of possible values (called sensitivity), the greater the amplitude of the added noise.

Definition 1 (*ϵ-local Differential Privacy*)**.** A random mechanism $\mathcal{M}$ satisfies ϵ-local differential privacy if, for any pair of input values v_1, v_2 in the domain of $\mathcal{M}$ and any possible output y of $\mathcal{M}$:

$$\Pr[\mathcal{M}(v_1) = y] \leq e^{\epsilon} \cdot \Pr[\mathcal{M}(v_2) = y]. \tag{1}$$

LDP mechanisms (Yang, Lyu, Zhao, Zhu, & Lam, 2020) are calibrated based on the data types they handle, which include real and integer values, among others. They are tailored to address various questions, such as frequency estimation, the identification of heavy hitters, and joint distribution analysis. One prominent mechanism among these is the Laplace mechanism, which is directly applicable to numerical data.

Definition 2 (*Laplace Mechanism in an Interval of Amplitude Δ*)**.** In the Laplacian mechanism, a numerical value v in an interval of amplitude Δ is sanitized into a numerical value

$$\mathcal{M}_{\text{Lap}}(v, \Delta, \epsilon) = v + \text{Lap}\left(\frac{\Delta}{\epsilon}\right) \tag{2}$$

where $\text{Lap}\left(\frac{\Delta}{\epsilon}\right)$ is the Laplace distribution centered in 0 and whose scale parameter is $\frac{\Delta}{\epsilon}$.

Laplace mechanism is however not adapted if it is required to return a precise answer (*i.e.* without noise, location name for instance). One solution is the exponential mechanism (McSherry & Talwar, 2007) recalled here which is going to select the sanitized output from a provided set element whilst preserving ϵ-local differential privacy.

Definition 3 (*Exponential Mechanism McSherry & Talwar, 2007*)**.** Let v in a domain D be the value to sanitized, $\mathcal{R}$ be the set of possible output sanitized data and $U : D \times \mathcal{R} \to \mathbb{R}^+$ be a scoring function with sensitivity Δ_U. The *exponential mechanism* sanitizes v to r with probability proportional to $\exp \frac{\epsilon U(v,r)}{2\Delta_U}$.

Table 3
Comparison of NER metric results across diverse entities. The highest scores in any language (resp in French language only) for each entity with respect to Precision, Recall and F_1 metric are marked in **bold**, (resp. in *italic*). NA stands for not addressed entity.

Method	CamemBERT-ner			MEDINA			FlauBERT-ner			Tchouka et al. (2022)			Proposal			Dernoncourt et al. (2016)		
Dataset	HNFC															i2b2		
Metric	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
PER	89	99	93.8	***98.2***	97.7	*98.2*	91.8	97.6	94.6	96.3	***99.8***	98	97.2	98.9	98	**98.2**	99.1	**98.6**
ORG	7.	21.8	11.1	32.6	24.8	28.1	16.9	34.1	22.6	41.1	*57.3*	47.8	*90*	51	*65.6*	**92.9**	**71.4**	**80.7**
LOC	46	67.2	54.6	98.8	81.1	89.1	75.7	66.3	70.7	88.4	***95.8***	92	***99.4***	94.4	***96.9***	95.9	95.7	95.8
DATE	NA			97.7	86.6	91.9	NA			97.7	86.7	91.9	***99.2***	*95.7*	*97.4*	99	**99.5**	**99.2**
AGE	NA			91.5	66.9	77.3	NA			91.5	66.9	77.3	*98.2*	*91.8*	*95*	**98.9**	**97.6**	**98.2**
TEL	NA			99.5	97.9	98.7	NA			***99.5***	97.9	98.7	99.4	***99.8***	***99.6***	98.7	99.7	99.2
REF	NA			NA			NA			NA			96.1	79.5	87	NA		
QID	NA			NA			NA			NA			77.2	32	45.3	**99.2**	**98.7**	**99**
Mic.-avg.	70.8	51.5	59.6	98.2	91.2	94.5	85.8	86.7	86.3	94.6	94.9	94.7	***98.5***	*96.4*	*97.4*	98.3	**98.5**	**98.4**

3.2.2. Date and age: Substitution strategy

Let us first recall that applying a uniform shifting on dates (Uzuner et al., 2007; Uzuner, Sibanda, Luo, & Szolovits, 2008) is definitively not secure. Indeed this shifting does not modify intervals between any temporal event and these ones are almost unique as shown in Tchouka et al. (2022).

In Tchouka et al. (2022), to maintain the chronology of events the following threefold approach is implemented. First, all temporal events of the textual document are normalized into $[e_0, \ldots, e_n]$ an ordered sequence of dates expressed as d-m-y, e_0 being the current date and e_n the oldest one. Next, the sequence $[e_0 - e_1, \ldots, e_{n-1} - e_n]$ of duration between consecutive dates is computed. The input domain is, therefore, $\mathbb{R}^+$ with the absolute value representing the distance. Then, the authors conclude that this is the segmentation of a dataset comprising dates into three distinct categories. These categories were defined on the basis of temporal spans, namely: less than 2 months, less than 2 years, and more than two years. The primary objective of this segmentation was to minimize sensitivity, which corresponds to the level of noise introduced in the data. However, it is important to note that even within the larger temporal category (more than two years), the amount of introduced noise remained substantial. This is due to the necessity of ensuring the indistinguishability of sanitized dates with a considerable difference in temporal spans, such as 3 years and 80 years.

Therefore, it becomes imperative to conduct a more detailed segmentation of the temporal space. Alternatively, one may also consider allowing distinctions between certain dates, particularly when the original temporal intervals are significantly apart. The core concept here revolves around the notion that a differential privacy mechanism should not forcibly equate two dates that exhibit substantial temporal disparity. Consequently, the level of privacy protection should depend on the distance or dissimilarity between the values of the elements requiring protection. This concept aligns with the framework of ϵ metric-privacy, as introduced in the work by Alvim, Chatzikokolakis, Palamidessi, and Pazii (2018), which is recalled below.

Definition 4 (ϵ *Metric-privacy*)**.** A randomized algorithm $\mathcal{A}$ satisfies the ϵ metric-privacy if, for any pair of input values v_1, v_2 in the domain of $\mathcal{M}$ with a metric d and any possible output y of $\mathcal{M}$:

$$\Pr[\mathcal{M}(v_1) = y] \leq e^{\epsilon . d(v_1, v_2)} \cdot \Pr[\mathcal{M}(v_2) = y]. \tag{3}$$

Instinctively, ϵ metric-privacy protects the accuracy of confidential information: by introducing a metric into the date space, it enables differentiation between an older date (like date of birth) and a recent date (such as the date of an operation the week before). Simultaneously, it ensures that two very recent dates (v_1 and v_2) at an extremely small distance will likely produce the same output y.

It has also been demonstrated (Chatzikokolakis, Andrés, Bordenabe, & Palamidessi, 2013) that the properties of composition and post-processing are possessed by an algorithm that satisfies the ϵ metric-privacy. We are then left to implement a mechanism ensuring this ϵ. metric-privacy property for temporal events. In the same article, it has been be further demonstrated that the mechanism that adds noise following $\text{Lap}\left(\frac{1}{\epsilon}\right)$ (i.e. the Laplace distribution centered in 0 and whose scale parameter is $\frac{1}{\epsilon}$) adheres to the ϵ metric-privacy property.

Property 1. *Guaranteeing ϵ metric-privacy by applying the Laplace mechanism is equivalent to guaranteeing a local Differential Privacy property with a Laplace mechanism whose sensitivity is reduced to 1.*

Proof. Let us consider the rate

$$\begin{aligned} \frac{\Pr[\mathcal{M}(v_1) = y]}{\Pr[\mathcal{M}(v_2) = y]} &= \frac{e^{\frac{-|y-v_1|}{b}}}{b} \times \frac{b}{e^{\frac{-|y-v_2|}{b}}} \text{ (where } b = \tfrac{\Delta}{\epsilon}) \\ &= e^{\frac{|y-v_2|-|y-v_1|}{b}} \\ &\leq e^{\frac{|v_2-v_1|\epsilon}{\Delta}} \end{aligned}$$

Achieving ϵ-metric privacy is thus equivalent to verifying Local Differential Privacy (LDP) when the sensitivity Δ is 1. □

In what follows, we consider the rate $\frac{\epsilon}{\Delta}$ to be equal to 1. This approach could potentially be perceived as a privacy vulnerability. In the context of ϵ-LDP, the Δ-amplitude associated with this mechanism would be fixed at 1 day, as opposed to the amplitude corresponding to individual categories, for instance, a substantial range like 100×365 days for the largest category. Consequently, specific dates, such as birthdays or medical intervention dates, are prone to substantial modifications. In contrast, age spans covering several decades are less likely to be altered, thereby giving unsatisfactory results. This issue was addressed as follows. First, it is worth noting the following observation: in a medical document containing a statement such as "10 years ago", the intended interpretation is more likely to be "approximately 10 years ago" rather than "on the exact same day 10 years ago". This type of approximation is also applicable when temporal events are expressed in terms of months or weeks. The metric under consideration becomes unit-dependent, employing years (or months, weeks, etc.) for events expressed in the corresponding units. As a result of this adaptation, an individual's age (measured in years) is expected to undergo a modification by a few years, for instance.

It is worth noting that, similar to classical differential privacy approaches, the privacy global ϵ budget is shared among all the elements to be substituted. This sharing can be either uniform or not, and in the absence of specific details, it is assumed to be uniform in this context.

Thread Example. In our thread example, there are 2 detected dates (expressed in days), 1 age expressed in years, and 1 location (duplicated twice), *i.e.* 4 contributions to substitute. Due to the sequential composition theorem (Kairouz, Oh, & Viswanath, 2015) of differential privacy, the ϵ budget must be distributed among the various contributions of an individual. Each element will thus consume $\frac{\epsilon}{4}$ of the privacy budget,

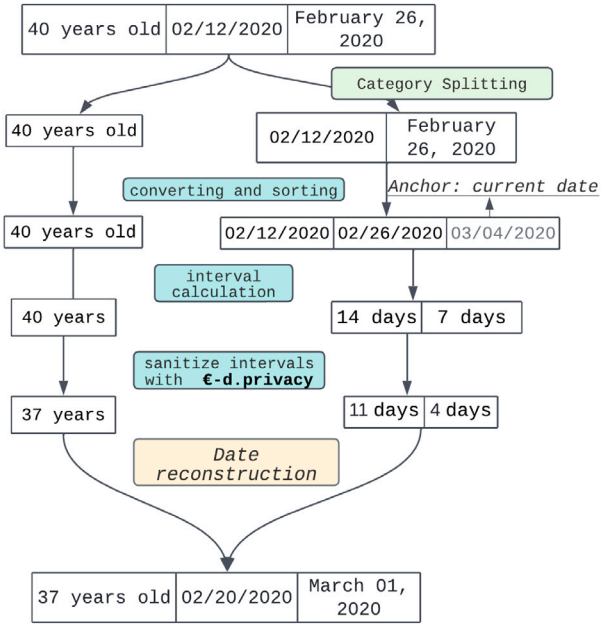


Fig. 6. Example of date substitution process on the thread example.

the last quarter is dedicated to sanitizing location. The date substitution process for this example is detailed in Fig. 6. Thus, **40** years, whose unit is day, and thus whose metric is expressed in years becomes 37 years. Similarly **02/12/2020** and **February 26, 2020** whose unit, and thus whose metric is expressed in days respectively lead to 02/20/2020 and March 01, 2020.

3.2.3. Geographic locations: Substitution strategy

Geo-indistinguishability (Bordenabe et al., 2014) is widely accepted as the de facto gold standard (Fawaz & Shin, 2014; Xiao & Xiong, 2015) to preserve location privacy. This mechanism effectively realizes ϵ metric-privacy (as recalled in Definition 4) in the context of locations, which are represented as (x, y) coordinates within $\mathbb{R}^2$.

In the de-identification context, as outlined in Tchouka et al. (2022), the authors employ geo-indistinguishability to introduce random noise to the coordinates (x, y) of the location to be sanitized, resulting in a new tuple (x', y'). Subsequently, they re-associate the nearest location to this new tuple. This method successfully safeguards privacy and provides a coherent substitute in the document. However, we contend that it may not fully address the question of the document's utility in a medical context. The concern stems from the fact that two locations that are geographically close to each other may be far apart from a health perspective. For instance, what is the In an example, this could lead to substituting the city of Dijon with the village of Beze, located at a distance of 23 km. However, with the aim of preserving the medical utility of the data, what is the relevance of this substitution when considering that the former has approximately 160,000 inhabitants and rates of cancer and stroke at 182 and 174, respectively, while the latter has about 700 inhabitants with cancer and stroke rates approximately equal to 1?

When substituting locations, it is desirable to randomly select among locations that are not only close geographically but also close in a statistical sense, considering factors such as population size, and in a medical sense, considering variables such as the incidence rate of all cancers, the number of strokes, air pollution, radon levels, etc. The key here is to express a distance between locations that incorporate statistical and medical features. Fortunately, many institutional websites provide this local information freely.[4][5]

Table 4, within its blue-framed section, provides an excerpt for some cities in France. With features available for each location, it is straightforward to calculate the distance between them, such as the Euclidean distance, and then apply any Local Differential Privacy (LDP) mechanism capable of capturing this distance.

For instance, let us consider a public database $\mathcal{R}$ of N locations, where each location i in $\mathcal{R}$, is represented as a $n+2$ length vector $(x_i, y_i, c_i^1, \dots, c_i^n)$, with (x_i, y_i) being the geographical coordinates, and $(c_i^1, \dots, c_i^n)$ representing features, which are normalized within the range $[0, 1]$. Let d_{ji} be the norm of feature differences between the two locations j and i, *i.e.* $d_{ji} = \|(c_i^1 - c_j^1, \dots, c_i^n - c_j^n)\|_2 \in \left[0, \sqrt{n}\right]$.

Let $v_j = [(1, d_{j1}), (2, d_{j2}), \dots, (j, 0), \dots, (N, d_{jN})]$ be the sequence of all distances between j and the other locations. In a practical situation, this sequence can be reduced to the location distances (i, d_{ji}) such that the geographical distance between i and j is lower than a given threshold and the k smallest values of the distances, where values are sorted in ascending order according to d. The sequence $v'_j = [(i_1, d_{ji_1}), \dots, (i_k, d_{ji_k})]$ is this result. The set $\mathcal{R}_j = \{i_1, \dots, i_k\}$ constitutes the possible substitutes for the city j. Notice that $(i_1, d_{ji_1}) = (j, 0)$ since the smallest distance is 0 between j and itself. The retained score function $U : \mathcal{R} \times \mathcal{R} \to \mathbb{R}^+$ is defined by

$$U(j, i) = \begin{cases} 1 - \frac{d_{ji}}{\sqrt{n}} & \text{if } i \in \mathcal{R}_j \\ -\infty & \text{otherwise} \end{cases}$$

Notice that $\sqrt{n}$ is a normalization factor allowing the score to belong to $[0, 1]$. This function is public and is not based on any private data. The probability distribution function is thus as follows:

$$P_j = [a.e^{\epsilon U(j,i_1)}, \dots, a.e^{\epsilon U(j,i_k)}, 0, \dots, 0] \tag{4}$$

where $a = \left(\sum_{i=1}^{k} e^{\epsilon U(j,i)}\right)^{-1}$ is the normalization factor. Notice this mechanism is an adaptation of the centralized exponential mechanism with public data, i.e., without sensitivity. The next section shows it verifies ϵ metric-privacy.

Property 2. *The mechanism defined in previous section verifies ϵ metric-privacy.*

Proof. According to Definition 4, for any y whose probability distribution definition is not null we successively have

$$\begin{aligned} \frac{\Pr[\mathcal{A}(v_1)=y]}{Pr[\mathcal{A}(v_2)=y]} &= \frac{ae^{\epsilon U(v_1,y)}}{ae^{\epsilon U(v_2,y)}} = \frac{e^{\epsilon(1-d(v1,y))}}{e^{\epsilon(1-d(v2,y))}} \\ &= e^{\epsilon(d(v_2,y)-d(v_1,y))} \leq e^{\epsilon.d(v_1,v_2)} \quad \square \end{aligned}$$

Clearly, the functionalities to be integrated for this step must be defined upstream in collaboration between the medical teams (who know the data) and the technical teams.

Thread Example. Let us take "Dijon" as an example and consider characteristics such as the overall population, cancer incidence rate, and strokes (as shown in blue in Table 4) for a database of all cities in France. The columns ('distance' and 'scores') represent, respectively, the vector distance (Euclidean distance with normalized features) and the results of the score function U from Dijon to $k = 10$ 'nearby' cities (according to features). After applying the probability distribution function previously detailed, we obtain the normalized distribution illustrated in orange in Table 4. The random draw thus follows this distribution.

[4] https://geodes.santepubliquefrance.fr/.
[5] https://www.insee.fr/.

Table 4
Example input geographical and epidemiological data used in the sanitization of Dijon city.

city	overall population	stroke	cancer incidence rate	distances	scores	normalized distribution
DIJON	160204	273.184785	182.252004	0.000000	1.000000	0.117964
BESANCON	119249	218.375283	134.135495	0.347525	0.799356	0.112193
CHALON SUR SAONE	46603	108.706972	52.730489	1.042888	0.397888	0.101479
DOLE	24606	55.290112	57.437117	1.381583	0.202343	0.096637
LE CREUSOT	21935	51.165964	24.819073	1.407732	0.187245	0.096273
MONTCEAU LES MINES	18789	43.827550	21.259429	1.454262	0.160381	0.095629
LONS LE SAUNIER	18023	40.497996	42.070599	1.475374	0.148193	0.095338
BEAUNE	21747	37.083653	24.739921	1.497023	0.135694	0.095041
AUTUN	14381	33.545372	16.271853	1.519458	0.122741	0.094733
VESOUL	15728	33.302482	42.069461	1.520998	0.121852	0.094712

According to memorization, all occurrences of the location **Dijon** can be replaced by **Besançon**.

Therefore, the final result of the substitution step would be: "**Mr. Julien, born in Besançon, 37 years old, was admitted to the hospital from 02/20/2020 to March 01, 2020 following a road accident in Besançon**".

4. Automatic ICD-10 code association system

This section presents the ICD-10 automatic code association task and outlines the challenges it poses. Within this context, we present our proposed architecture that address these challenges and the experiments that were conducted.

4.1. ICD-10 code association task

To ensure accurate long-term follow-up, patient details during their stay in a healthcare facility are typically documented in digital records, constituting the patient's medical record. These records, which include operative reports, clinical notes, medical correspondence, and other elements, are written by the attending physicians. In many countries, each patient's file is then categorized according to the International Classification of Diseases (ICD), a medical classification system managed by the World Health Organization (WHO) and globally adopted for encoding diseases and other health conditions. This study focuses on the 10th edition of the ICD (ICD-10) (Organization et al., 1992). Automatically associating ICD-10 codes with medical records has emerged as a significant research challenge in the contemporary medical scientific community (Baumel et al., 2018; Choi et al., 2016; Dalloux et al., 2020; Huang et al., 2022; Vu et al., 2020).

4.2. Challenges

ICD coding involves assigning a set of codes to a given medical record, constituting a multi-label text classification task. However, creating an efficient model to automate this ICD code association is a complex undertaking.

One primary challenge arises from the extensive number of ICD-10 codes, totaling around 140,000 distinct codes, encompassing both procedure and medical codes. Achieving high accuracy by associating any of the 140,000 existing codes with a medical record is unrealistic without access to a massive dataset, significant resources, and considerable time. In practice, ICD-10 code association datasets are notably smaller compared to the comprehensive ICD-10 code set. For instance, the English MIMIC-II and MIMIC-III corpora contain 5031 and 8922 different codes (labels), respectively. The substantial number of labels presents a formidable challenge for conventional classification models.

Another significant challenge stems from the size of medical records typically subject to ICD code association. Medical notes often exceed the sequence limit that Transformer architectures can handle, typically set at 512 words. As depicted in Table 2, the average size of medical notes in this study's dataset is approximately 747 words, surpassing the capacity of conventional Transformer (encoders) models.

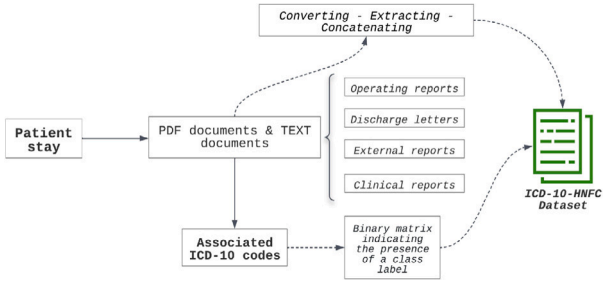


Fig. 7. ICD-10-HNFC dataset construction.

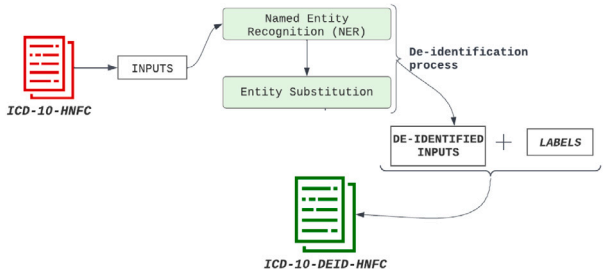


Fig. 8. ICD-10-DEID-HNFC dataset construction.

4.3. Dataset

A patient's stay comprises a series of successive visits to possibly different departments within the hospital. Each department generates a clinical document detailing the patient's experience during their stay in that specific department. These clinical documents are used by medical coding specialists to associate the relevant ICD-10 codes. Consequently, a collection of unstructured textual documents is obtained that correspond to the comprehensive patient stay, each associated with a set of codes. These clinical documents encompass various types, such as operating reports, discharge letters, external reports, or clinical notes. The resulting dataset, referred to as the ICD-10-HNFC dataset, thus constitutes a repository of groups of medical documents each linked to associated codes. This system is best illustrated in Fig. 7.

However, the ICD-10-HNFC dataset remains a private dataset containing sensitive information about individuals. In the interest of privacy, our whole de-identification approach, proposed in the previous section, was applied to this dataset to produce a de-identified dataset that we will refer to as ICD-10-DEID-HNFC. For all these ϵ-metric privacy based algorithms, the privacy parameter ϵ have been set to 1 as well as the value of Delta. The selection of the Δ value is guided by Property 1, while the choice of the ϵ value is a compromise between utility and privacy.

This process is illustrated in Fig. 8. The training process will then be experimented with on the ICD-10-DEID-HNFC dataset.

Table 5
Descriptive statistics of ICD-10-HNFC dataset.

	Dataset	Dataset with class reduction
Documents	56 000	–
Tokens	41 868 993	–
Average sequence length	747	–
Total ICD codes	416 125	415 830
Unique ICD codes	**6160**	**1564**
Codes with less than 10 ex.	3722	523
Codes with 100 ex. or more	641	471

Class reduction

Additionally, we delve into sub-dataset, incorporating the association of the code families. It addresses the challenge of large label sets by reducing the codes to the first 3 characters, treated as a family. Instead of considering raw codes individually, they are grouped into families. This substantial reduction in the number of classes to be handled by the model is evident in Table 5. The "Code with less than 10 examples" category in Table 5 demonstrates that this reduction not only provides a more manageable number of classes but also increases the frequency of codes in the dataset.

4.4. Model architecture

This section introduces the various components of the model architecture that was developed and provides justification for the choices made in its design. As previously mentioned, given that we are working with the French language, our approach involves fine-tuning pre-trained transformer-based French models, specifically CamemBERT (Martin et al., 2020) and FlauBERT, to implement the model architecture.

4.4.1. Global document representation

As previously highlighted, Transformers face a constraint regarding the maximum number of tokens present in an input sequence. Given that the average size of clinical notes in the ICD-10-HNFC dataset surpasses this limit (747 versus 512, as shown in Table 5), conventional Transformers become impractical. Recently, Dai, Chalkidis, Darkner, and Elliott (2022) provided a summary of available methods to process long sequences using Transformers, which includes hierarchical Transformers and sparse-attention Transformers. Notably, the *Longformer* model by Beltagy et al. (2020) falls into the category of sparse-attention Transformers and can process up to 4096 tokens per sequence, thereby overcoming this limitation.

However, as of now, there is no pre-trained *Longformer* model available for the French language. Consequently, in this paper, the hierarchical method was chosen to address the challenge posed by the length of clinical notes in the dataset.

Hierarchical Transformers (Dai et al., 2022; Pappagari et al., 2019) are an extension of the Transformers architecture. In this approach, a document D is initially divided into segments $[t_0, t_1, \ldots, t_{|D|}]$, each containing fewer than 512 tokens (the limit of Transformers). These segments are then encoded independently using a pre-trained Transformers model. Consequently, we obtain a list of segment representations that must be aggregated to form the complete document representation for D. Various aggregation methods can be used for this purpose. The aggregator may compute the average of the representations of all the segments in the document (mean pooling), take the maximum value of the representations in each dimension across segments (max pooling), or stack the segment representations into a single sequence. The resulting aggregated sequence serves as input to the subsequent layer in the model.

Classification of a large number of labels

To address the challenge of dealing with a large set of labels, given that ICD-10-HNFC contains more than 6000 codes, we used the Label-Aware Attention (LAAT) system introduced by Vu et al. (2020), similar to the approach in Huang et al. (2022). LAAT involves incorporating the labels into the document representation, allowing it to capture essential text fragments associated with specific labels.

Let H denote the stacked representation of an input sequence. Initially, a label-wise attention weight matrix Z is computed as follows:

$$Z = tanh(VH)$$

$$A = softmax(WZ)$$

where V and W are linear transforms. The ith row of A represents the weights of the ith label. The softmax function is performed for each label to form a distribution over all tokens. Then, the matrix A is used to perform a weighted-sum of H to compute the label-specific document representation:

$$D = HA^T$$

The ith row of D shows the document representations for the ith label. Finally, D is used to make predictions by computing the inner product between each row of D and the related label vector.

In this paper, several architectures were experimented with, including the model without long sequence processing, the model with long sequence processing using max/mean pooling, and the model with Label-Aware Attention (LAAT). The overall architecture is illustrated in Fig. 9.

5. Effect of de-identification on medical utility

In this section, we will evaluate the preservation of the usefulness of the de-identification method presented in Section 3.

To measure the usefulness of de-identification, we used the machine learning system we just addressed in Section 4: ICD-10 code association. This task is relevant to measure the usefulness of de-identification, because the data used not only contain the sensitive information one needs to de-identify, such as names, geographical locations, ages, dates, etc., but also some of this information may influence the medical analysis of the document (e.g. the patient's age may direct the analysis towards a certain code category, or patients from the same location marked by a natural factor may present the same pathologies, etc.). The usefulness of de-identification will therefore be measured by the association performance or errors with a learning technique before and after the de-identification process. The construction of the datasets is detailed below.

5.1. Methodology

This section presents the methodological approach used to evaluate the usefulness of de-identification strategies. It starts with an introduction to the de-identification strategies used to build the various datasets. Then, the machine learning system that facilitated the evaluation of these different datasets is presented.

5.1.1. De-identified datasets ICD-10-DEID-HNFC and ICD-10-TAG-HNFC

We consider the original data, represented by the dataset ICD-10-HNFC, which is described in Section 4.3. This dataset is the input to generate the de-identified dataset ICD-10-DEID-HNFC following differential privacy based methods presented in Section 3.2 aiming at establishing our automatic association system for ICD-10 codes. Notice that as previously justified, the rate $\frac{\epsilon}{\Delta}$ is 1. The process of creating this de-identified datasets is illustrated in Fig. 8.

A second dataset ICD-10-TAG-HNFC has been generated by replacing each detected entity in ICD-10-HNFC by its associated entity label:

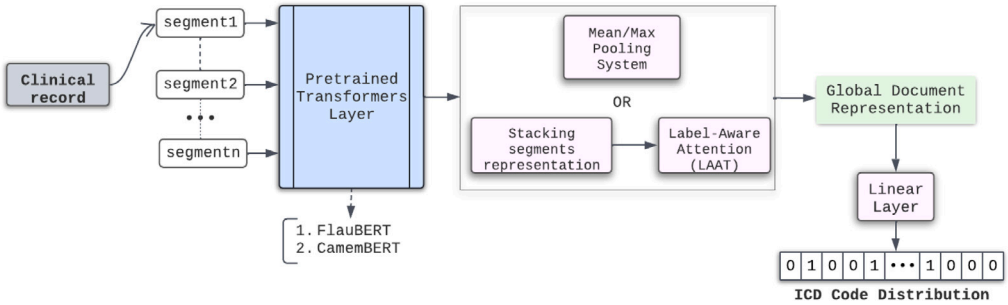


Fig. 9. Architecture of ICD-10 code association system.

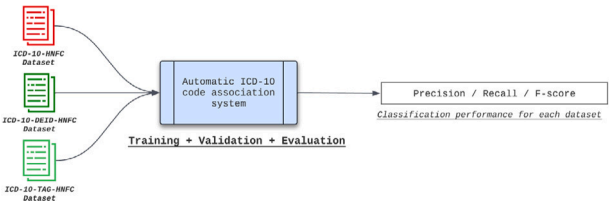


Fig. 10. Approach to evaluate the usefulness of de-identification.

Table 6
Evaluation of different datasets on the same validation dataset.

Dataset	Labels	Precision	Recall	F_1-score
ICD-10-HNFC		**0.47**	**0.46**	**0.47**
ICD-10-DEID-HNFC	**6160**	0.44	0.43	0.44
ICD-10-TAG-HNFC		0.43	0.41	0.42

the city name "DIJON" in the document is thus replaced by the entity "LOC" (for location). This de-identification approach is the most robust against potential re-identification attacks. However, by removing the values of the entities, they cannot be exploited for classification purposes.

5.1.2. ICD-10 code association system

After creating the various datasets, we implemented the ICD-10 code association system inspired by Huang et al. (2022) based on the architecture detailed in Section 4.4 to address the challenges mentioned earlier. The supervised learning process (including training, validation, and evaluation), is then applied to the three datasets: ICD-10-HNFC, ICD-10-DEID-HNFC, and ICD-10-TAG-HNFC. This process is illustrated in Fig. 10.

6. Experiments

This section presents the evaluations of the paper. Initially, we measure the usefulness of de-identification with a CIM code association system inspired by Huang et al. (2022) on datasets with various levels of de-identification. Then, we present the results of different architectures developed within the scope of this work on the original dataset.

The evaluation of our models is performed using commonly used classification performance measures: Precision, Recall, and F_1-score. The micro-average system is used to aggregate the performances.

6.1. Experiments on the usefulness of de-identification

Regarding the evaluation, we have chosen the main task, which involves classifying all the codes present in the dataset (6160 labels, as indicated in the dataset's descriptive table in 5). We apply the same performance criteria as those described above, namely precision, recall, and the F_1 score. The experiments conducted on the three datasets with the same instance of input data using the same parameters. The results obtained are presented in Table 6.

We observe that the results obtained in the evaluations show disparities among the datasets, despite using the same learning process and the same validation data. These variations confirm, even without taking the results into account, that de-identification has an impact on the medical document analysis, particularly with regard to the association of ICD-10 codes.

The highest performance among the three datasets is obtained from the evaluation of the original corpus ICD-10-HNFC. This illustrates how the de-identification process leads to a deterioration of the document's usefulness. The evaluation on the ICD-10-TAG-HNFC corpus resulted in the lowest F_1 score. In view of the evaluations carried out on the ICD-10-DEID-HNFC and ICD-10-TAG-HNFC corpora, which correspond respectively to the de-identification proposed in this paper and to partial de-identification, it is clear that partial de-identification has the most negative impact on the document's usefulness. The percentage gap between the original corpus ICD-10-HNFC and the de-identified corpus ICD-10-TAG-HNFC is approximately 12%. The de-identification process described in Section 3 reduces this gap to 6.8%. This reinforces the idea that the question of medical utility in the context of de-identification is perfectly justified.

6.2. Experiments on original dataset: ICD-10-HNFC

In this section, we present the experiments carried out on the original dataset using various architectures. We compare the outcomes with recent works, such as PLM-ICD (Huang et al., 2022) and CNN (Dalloux et al., 2020).

6.2.1. Models

Our initial experiments were conducted on the ICD-10-HNFC dataset with class reduction (1564 labels), as outlined in Fig. 7. These experiments encompassed all the architectures developed in this paper, as listed in Table 7. Subsequently, we trained another model on the complete ICD-10-HNFC dataset (6160 labels) using the architectures that achieved the highest F_1-score in the previous experiment. The results are presented in Table 7. These findings confirm the impact of the different components within our architectures. In summary, the LAAT approach outperforms the hierarchical methods, which, in turn, outperform the basic truncated model.

6.2.2. Analysis

Table 8 showcases the model with the highest F_1-score from this paper alongside the results of previous work on ICD-10 code association. Comparing these results is challenging due to the variation in evaluation datasets, and English works may benefit from specialized models such as ClinicalBERT (Alsentzer et al., 2019). As a French baseline, we implemented and trained the model proposed in Dalloux

Table 7
ICD-10 association results of the different architectures on the validation ICD-10-HNFC dataset.

Models	Labels	Precision	Recall	F_1-score
FlauBERT (512 tokens)	1564	0.48	0.31	0.38
Hierarchical mean FlauBERT		0.54	0.39	0.45
Hierarchical max FlauBERT		0.53	0.40	0.46
FlauBERT + LAAT		**0.57**	0.51	0.54
CamemBERT + LAAT		0.56	**0.53**	**0.55**
FlauBERT + LAAT	6160	0.41	0.43	0.42
CamemBERT + LAAT		0.52	0.4	**0.45**

Table 8
Results comparison with the previous work on ICD-10 Association. The state of the art works with their results are in *italic*. The experiments carried out in this paper with ICD-10-HNFC dataset are presented in the other part. The highest scores in each part in relation to the number of labels are marked in **bold**.

Models	Language	Dataset	Labels	F_1-score
PLM-ICD	*English*	*MIMIC 2*	*5031*	*0.5*
Huang et al. (2022)		*MIMIC 3*	*8922*	***0.59***
Dalloux et al. (2020)	*French*	Dalloux et al. (2020)	*6116*	*0.39*
			1549	*0.52*
Proposal	French	ICD-10-HNFC	6160	**0.45**
			1564	**0.55**
Dalloux et al. (2020)			6160	0.27
			1564	0.35

et al. (2020) on the ICD-10-HNFC dataset. The results are juxtaposed with our proposal.

Our model significantly outperforms the classification method used in Dalloux et al. (2020). On the same validation dataset, with class reduction (1564 labels), the F_1-score improves from 0.35 (model proposed in Dalloux et al. (2020)) to 0.55 with our proposal, representing a 57% improvement. With the raw codes (6160 labels), the F_1-score improves from 0.27 to 0.45, indicating a 66.6% improvement. The variance in scores compared to PLM-ICD results may be attributed to the use of a context-specific (medical) Transformers with a vocabulary more tailored to the content of medical documents.

7. Discussion

An observation potentially raised regarding this research could pertain to the use of models deemed "outdated" in comparison to those relying on Large Language Models. Nevertheless, this can be mitigated by considering two fundamental points. Firstly, our study focused on the specific domain of medical French, for which, to the best of our knowledge, there exists no appropriate substantiation of LLM. Secondly, due to confidentiality imperatives, all stages of development and experimentation were conducted within the hospital site. This particularity resulted in a complexification of the process, given that the potentially accessible hardware significantly differs from that typically available in a conventional computing environment, and data access is subject to stringent regulations aimed at facilitating our approach. These constraints unquestionably extended the required duration for project completion, a consideration particularly detrimental in the current temporal context.

8. Conclusion

This paper addresses the de-identification of medical data for medical analyses within the framework of scientific research. Firstly, we tackle the de-identification task itself, which encompasses the Named Entity Recognition (NER) task and the generation of substitutes. For the Named Entity Recognition, an existing, comprehensive but imperfect de-identification approach was used to internally build a new and substantial dataset of medical records (approximately 15,000 sentences). Leveraging this large labeled dataset, deep learning was implemented with a Transformer-based architecture. The results obtained represent the state-of-the-art in the French language and are as precise as those achieved in English.

To substitute the detected sensitive information, we integrate the notion of medical utility through the differential privacy mechanism based on a metric (ϵ metric-privacy), mainly at the level of attributes that can impact medical analysis, such as temporal data (dates and ages) and geographical locations. To evaluate the proposed de-identification strategies, we place this work in a direct application context, which is the association of ICD-10 codes.

We start by addressing the scientific challenges of ICD-10 code association, which is a multi-label text classification task, namely the significant number of ICD-10 codes and the size of medical data. In this paper, we propose the most accurate model to date for the automatic association of ICD-10 codes in the French language. Subsequently, we used this machine learning system to evaluate our de-identification approach. This evaluation confirmed the preservation of utility in our approach compared to conventional de-identification methods (e.g., substituting attributes with their labels).

Even if the presented approach meets the requirements regarding privacy protection by incorporating differential privacy, resulting datasets are not immune from a re-identification or membership attack on the dataset. Therefore, even de-identified datasets cannot be disseminated. The question of the risk of re-identification of de-identified documents remains complex. An in-depth analysis of vulnerabilities and potential attacks against the algorithms proposed in this paper is necessary an is planed as a future work to provide a more precise answer to this question.

Finally, hospitals publishing ICD-10 codes generate temporal statistics for institutions and receive funding based on care provided, with codes serving as a summarized version. The objective can thus be seen as more medically administrative than purely medical. On the opposite, using personal data, even de-identified ones, in medical research falls under the application of the French law Jardé,[6] which is much stricter and takes longer to implement. The theoretical and practical approaches taken in this article, both in de-identification and classification, can also be helpful in such contexts.

This study empirically demonstrated that the utility of the de-identification process decreased by 12% (in terms of a classification score) when replacing each entity with its TAG. It decreased by 6.8% when using an ϵ-LDP approach, where the ratio $b = \frac{\epsilon}{\Delta}$ is 1. Formally, substituting an entity with its TAG corresponds to de-identification with $b = 0$, and not modifying the attributes corresponds to $b = \infty$. Between these extreme values, we plan to evaluate how the classification score evolves as a function of the ratio b. As an another application, in future work, we plan to detect treatment pathways disrupted by healthcare-associated infections

CRediT authorship contribution statement

Yakini Tchouka: Conceptualization, Writing – original draft, Writing – review & editing. **Jean-François Couchot:** Supervision, Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **David Laiymani:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Philippe Selles:** Supervision, Validation. **Azzedine Rahmani:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

[6] .

Data availability

The data that has been used is confidential.

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